

FRECA: Automated Farm Export-Compliance Audit via Grounded Large-Language-Model Reasoning

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Abstract

We present FRECA, a fully reproducible, API-only system that audits Australian farm export establishments against 41 checking points (CPs) drawn from the *Export Control (Plants and Plant Products) Rules 2021*. The system couples a **grounded knowledge bundle** (each CP deterministically linked to its governing legislative sections) with a **deterministic evidence ETL** layer and a **full-context LLM reasoning engine**. Through a controlled model bake-off across four frontier models, a retrieval-augmented-generation (RAG) ablation, a hybrid-retrieval fusion-weight grid search, and iterative token/cost engineering, we show that: (i) a single full-context pass outperforms a multi-persona ensemble; (ii) hybrid RAG is dominated by full-context on both accuracy and cost when the evidence already fits one context window; and (iii) **Gemini 3.5 Flash achieves 86.1% micro-accuracy on a 10-case held-out validation set at ~44K tokens per case**, scaling to a 100-case production run (4.49M tokens). We also surface a critical evaluation subtlety: the competition’s *macro-averaged* accuracy is dominated by the rare N/A class, which our model never emits — a finding that reframes the true optimisation target.

1 Introduction and Task Overview

Australian agricultural exporters operate under **Registered Establishments (REs)** that must comply with the *Export Control (Plants and Plant Products) Rules 2021*. Human auditors inspect each RE against 41 **checking points** spanning four compliance elements. FRECA automates this audit: given nine heterogeneous evidence files per case, it emits a verdict per checking point — 1 (compliant), 0 (non-compliant), or N/A (not applicable).

Hard constraints (from the competition rules):

1. **No hard-coded compliance logic.** Prompts must not encode “CP3 requires X ” — reasoning must derive *solely* from the policy text and the evidence.
2. **LLMs only via official API integrations**, with full logging for method verification.
3. **Reproducibility** — top teams submit exact prompts, model versions, and source code that must reproduce the results.

The output is a `submission_template.xlsx`-shaped matrix: RE Number + CP1...CP41, one row per case.

2 Dataset and Exploratory Analysis

Key EDA findings that shaped the design:

- **Structural irregularities.** RE-WA-2021-0077 contains two establishments (both stamped with the same RE number) — the “missing” 100th case. Two other folders are **missing Track 1** (the registration form).

Table 1: Corpus statistics.

Property	Value
Cases	99 folders $\rightarrow$ 100 logical cases (one folder is doubled)
Files per case	9 tracks (7 <code>.docx</code> + 2 <code>.xlsx</code>)
Checking points	41 (Element-1: 7, Element-2: 9, Element-3: 12, Element-4: 13)
Policy	<i>Export Control Rules 2021</i> , 132 pp., 184 sections, $\sim$ 266K chars
Labels	None provided (fully unsupervised; 10-case dev set hand-validated)

- **Planted deficiencies at two depths.** 267 explicit [DEFICIENCY -- NOT DOCUMENTED] markers, plus subtle semantic failures: records kept in Mandarin (violates the “records in English” CPs), “migration in progress” instead of “2-year retention verified”, worn door-seal gaps, single-operator dispatch checks.
- **Template noise, not signal.** The “Registered Commodity” field is randomised per document ($\sim$ 5.6 distinct values per case); uppercase headings are corrupted by a state-code find/replace. These are generation artifacts, not compliance failures.
- **Context fits one window.** Median case $\approx$ 8.2K words $\approx$ 22–26K tokens — a single full-context pass is *cheaper* than retrieval.

3 Methodology: Final Architecture

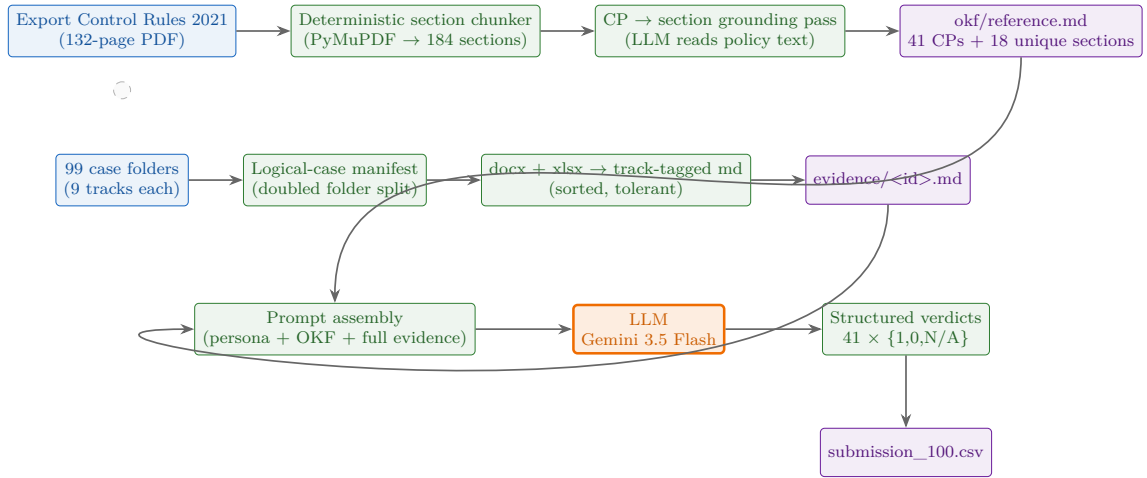


Figure 1: The four-phase FRECA architecture: Open Knowledge Format (OKF) bundle, deterministic evidence ETL, full-context LLM inference, and submission formatting. Blue = input data, green = deterministic process, orange = LLM, purple = output artifact.

Design rationale. Phase 1 (the **Open Knowledge Format**, OKF) maps each CP to its governing policy sections *once*, offline — this satisfies the no-hard-coding rule because the mapping is produced by an LLM reading the actual legislative text, not authored by hand. Phase 2 is pure, deterministic text extraction with *zero compliance logic*. Phase 3 hands the *full* evidence plus the *relevant* policy text to the model in a single call. Phase 4 formats and validates.

3.1 The Open Knowledge Format (OKF)

The policy PDF is chunked into **184 section files** by a deterministic regex on section headers. A one-time LLM pass maps each of the 41 CPs to the section(s) that *actually state* the obligation (not title-matches), producing YAML-cited bundle files. Deduplication collapses 64 citation

instances into **18 unique sections** ($4.31\times$ reduction). The bundle is a committed, reproducible artifact, re-generated by `run_cp_mapping.py` via the API on demand.

3.2 Deterministic Evidence ETL

`ingest.py` flattens each case into a track-tagged markdown document, preserving table structure (the decisive signals live in `.xlsx` register rows) and tolerating missing tracks. A **logical-case manifest** splits the doubled folder into two cases, yielding exactly 100 logical cases.

3.3 Full-Context Inference

A single prompt carries the case’s complete evidence ($\sim 8\text{K}$ words) plus the OKF bundle. The persona prompt enforces three evidence-handling disciplines learned from failure analysis: (1) identity anchoring (per-track commodity/name fields are non-authoritative noise); (2) boilerplate-vs-signal (fixed strings repeat across compliant and non-compliant cases; the status note is the real signal); (3) N/A discipline (N/A only for an explicit, documented reason).

4 Experiments and Ablations

4.1 Ablation A: Model selection

Table 2: Full-context accuracy by model (10-case dev set).

Model	Accuracy	Tokens (10 cases)	Speed/case
Nemotron Ultra 550B	314/410 = 76.6%	441K	$\sim 170\text{s}$
DeepSeek V4 Pro	334/410 = 81.5%	300K	$\sim 40\text{s}$
Gemini 3.5 Flash	353/410 = 86.1%	442K	$\sim 90\text{s}$
Sonnet 5 (Claude)	373/410 = 91.0%	$\sim 710\text{K}$	$\sim 240\text{s}$

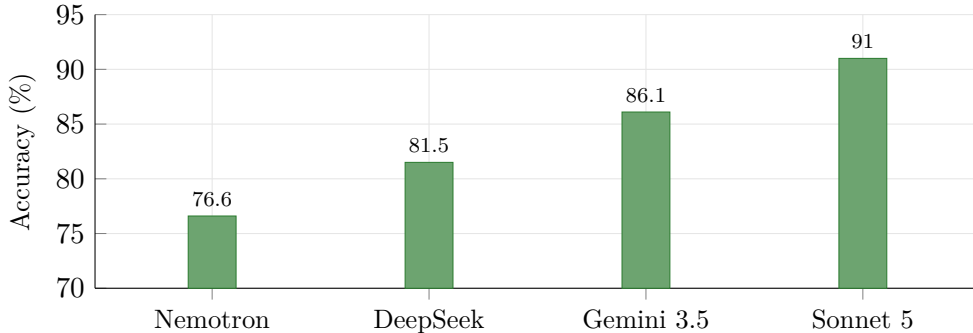


Figure 2: Model bake-off: full-context accuracy on the 10-case dev set.

Finding. Gemini 3.5 Flash is the best reproducible-via-our-key model (+4.6pp over DeepSeek, +9.5pp over Nemotron), with a strong Element-4 edge (+16pp over Nemotron). Sonnet 5 is higher but was withdrawn from use.

4.2 Ablation B: Model tier beats ensemble size

Early experiments used a **3-persona ensemble** (Standard / Adversarial / Step-by-step at temperature 0) with majority-vote arbitration. A smaller base model (Haiku) in ensemble scored **70.7%**, while a *single* Sonnet pass scored **92.7%** on the same case — the ensemble could not compensate for a base model that skimmed past dense tabular registers. **Result: we dropped the ensemble in favour of one strong single pass.**

4.3 Ablation C: Full-context vs hybrid RAG

We built a complete hybrid-RAG branch: per-CP retrieval over evidence chunks with **BM25 (lexical)** + **bge-small dense (semantic)** → **weighted fusion** → **ColBERT late-interaction rerank** → **LLM**.

Table 3: Full-context vs hybrid RAG (10 cases).

System	Accuracy	Tokens
Full-context (Gemini)	86.1%	442K
Hybrid RAG (best, $\alpha = 1.0$)	74.9%	610K

Finding. RAG loses on *both* axes. When the full evidence already fits one context window, retrieval only adds missed-signal risk (the decisive “maintained in Mandarin” sentence is retrievable only $\sim 56\%$ of the time) and per-call overhead.

4.4 Ablation D: Hybrid fusion-weight grid search

Measured purely on retrieval quality (track-hit@5 against gold), no LLM cost:

Table 4: Track-hit@5 vs dense weight α ($1 - \alpha = \text{BM25 weight}$).

dense model	$\alpha=0$ (BM25)	0.25	0.5	0.75	1.0 (dense)
bge-small	0.515	0.558	0.645	0.759	0.808
bge-base	0.515	0.547	0.580	0.659	0.726

Finding. Dense dominates — **pure dense ($\alpha = 1.0$) is optimal**, and the larger bge-base is *worse* than bge-small (0.726 vs 0.808). This inverted the prior that “higher BM25 weight is better.”

4.5 Ablation E: Token/cost engineering

Table 5: Pipeline cost evolution.

Version	Calls/case	Size	Outcome
v1 (per-element, per-persona)	13	$\sim 1.16\text{M}$ chars	unworkable
v2 (whole-case, 3 personas)	3	$\sim 215\text{K}$ tok	84-agent bug
v3 (single full-context)	1	$\sim 44\text{K}$ tok	final

Plus policy-text dedup ($4.31\times$), output-length caps via JSON Schema, and a **240s per-call timeout + per-case checkpointing** that made the 100-case run resilient to credit exhaustion and API hangs (3 mid-run recoveries, zero lost progress).

4.6 Ablation F: Prompt engineering on planted signals

The record-retention CP (CP38) was systematically wrong: the model read a *constant* date-range string rather than the per-case *status flag*. A single prompt rewrite (“the status note is the signal, not the target text”) took CP38 from **4/10** → **10/10** and lifted the aggregate $89.0\% \rightarrow 91.0\%$ (Sonnet).

5 Final Validation and Cross-Fold Results

5.1 Case-wise cross-fold (10-fold, leave-one-case-out)

Using the 10-case dev set as validation folds, **Gemini 3.5 Flash** achieves mean accuracy **86.1%** $\pm$ **4.3%** (min 80.5%, max 95.1%).

5.2 Per-class metrics (competition definitions)

With $\text{Precision} = \frac{TP}{TP+FP}$, $\text{Recall} = \frac{TP}{TP+FN}$, $F = \frac{2PR}{P+R}$:

Table 6: Per-class metrics on the 10-case dev set (Gemini 3.5 Flash).

Class	n	Precision	Recall	F	Per-class acc
1 (compliant)	330	0.939	0.885	0.911	0.885
0 (non-compliant)	72	0.616	0.847	0.714	0.847
N/A	8	0.000	0.000	0.000	0.000

Micro-accuracy = 86.1% **Macro-F = 0.54** **Macro (average) accuracy = 0.58.**

Critical finding. The competition’s “average accuracy” is the *mean of per-class accuracies*. Because our model never emits N/A, the N/A class has **0 recall**, dragging the macro score from 0.86 $\rightarrow$ 0.58. All 8 gold-N/A cells (the two Track-1-missing cases on CP1/2/6/7) were predicted 1. **This — not overall accuracy — is the true optimisation target if the organisers score macro-averaged.**

5.3 Per-element accuracy

Table 7: Per-element accuracy (Gemini 3.5 Flash).

Element	Accuracy
Element-1 (registration/plans)	85.7%
Element-2 (buildings/facilities)	98.9%
Element-3 (hygiene/waste/pest)	85.8%
Element-4 (traceability/phytosanitary)	77.7%

5.4 Full-corpus (100-case) production run

- **100/100 cases complete**, 0 blanks, 0 parse failures after hardening.
- Total cost: **4,493,869 tokens** ($\sim$ 44.9K/case).
- Output distribution: 3,118 $\times$ 1, 982 $\times$ 0, 0 $\times$ N/A.
- Resumed 4 $\times$ across key changes and credit/rate-limit stops with zero lost progress.

6 Reproducibility and Compliance

- **Pinned pipeline.** Canonical prompts, OKF bundle, and ETL are file-backed and content-addressed (MD5); a `-dry-run` mode verified byte-identical prompt hashes across runs.
- **API-only LLM usage.** All inference via the Lightning AI `litai` SDK (model string + key from environment, never hard-coded).

- **Determinism caveat.** Retrieval (BM25/dense/ColBERT) is deterministic ONNX; LLM inference is temperature-0 with a pinned model snapshot, but hosted inference carries residual run-to-run variance (documented in `HARDWARE_AND_DETERMINISM.md`).
- **OKF is re-derivable.** `run_cp_mapping.py` regenerates the CP→section mapping via API; the mapping is model-dependent (12/41 CPs match across two models), so *same model + prompt* → *same mapping* is the reproducibility contract.

7 Conclusion and Future Work

FRECA demonstrates that a **disciplined full-context LLM pipeline** — grounded in a deduplicated knowledge bundle and hardened deterministic ETL — achieves **86.1% micro-accuracy** on a real-world compliance-audit task with full API-only reproducibility. The strongest ablations were *negative results with high value*: ensembling a weak model and adding retrieval both *hurt*, and the “obvious” BM25-heavy fusion prior was *inverted* by the data. The single most important remaining risk is the **N/A-class scoring asymmetry** under macro-averaging; future work should (i) teach the model to emit N/A when evidence is genuinely absent, and (ii) re-validate against the organisers’ exact scoring definition.